

BIOS 755: Linear Longitudinal Workflow

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Longitudinal Continuous Outcomes: Modeling Roadmap

1. Study design and scientific question
2. Exploratory data analysis
3. Specify mean structure
4. **Modeling choice:**
 - ▶ Covariance Pattern Model (marginal)
 - ▶ Linear Mixed Model (random effects)
5. Model diagnostics
6. Interpretation

Core question: Are we modeling correlation directly, or via subject-specific random effects?

Step 1: Study Design Considerations

Design influences model choice.

- ▶ What is the exposure?
- ▶ How do we test the hypothesis of interest?
- ▶ What is the most biologically relevant time scale?
- ▶ Are measurements equally spaced or designed to be equal across subjects?
- ▶ How many repeated measures per subject?

Step 2: Exploratory Data Analysis

- ▶ Spaghetti plots with mean profiles by group
 - ▶ use at most ~ 1000 subjects for these (maybe less)
 - ▶ spaghetti plots are less useful with multiple continuous variables.
- ▶ Variance by time
- ▶ Missing data patterns

Questions raised by EDA:

- ▶ Do subjects differ in baseline level?
- ▶ Do subjects differ in slopes?
- ▶ Is the variance equal across time?
- ▶ How to represent the mean over time?

Step 3: Specify the Mean Structure

► Example:

$$E(Y_{it}) = \beta_0 + \beta_1 A_i + \beta_2 time_{it} + \beta_3 (A_i \times time_{it}) + Confounders$$

► Consider:

- Functional form of time (linear, quadratic, profile, spline)
- Confounders (baseline vs time-varying)

Mean model is common to both modeling approaches.

Step 4: Covariance Pattern Model

- ▶ Models:

$$Y_i = X_i\beta + \varepsilon_i \quad \text{with } \varepsilon_i \sim N(0, \Sigma_i)$$

- ▶ Directly specify Σ_i :

- ▶ CS
- ▶ CSH
- ▶ AR(1)
- ▶ UN

- ▶ Correlation handled through patterned covariance.

Strength: Flexible correlation modeling. **Limitation:** No subject-specific trajectory modeling. Most useful when observation times are equal across subjects.

Step 4: Linear Mixed Model

- ▶ Model:

$$Y_i = X_i\beta + Z_ib_i + e_i$$

- ▶ $b_i \sim N(0, G)$ (random effects)
- ▶ $e_i \sim N(0, R_i)$
- ▶ Correlation induced by shared random effects.

Key additional decisions:

- ▶ Random intercept only?
- ▶ Random slope?

Model Diagnostics: Pattern vs Mixed

- ▶ In covariance pattern models:
 - ▶ Check residual plots (use scaled residuals)
 - ▶ Compare covariance structures via AIC/BIC
- ▶ In linear mixed models:
 - ▶ Check conditional studentized residuals
 - ▶ Check marginal variance of Y
- ▶ In both:
 - ▶ Assess influential subjects
 - ▶ Perform sensitivity analyses

Putting It All Together

- ▶ Start with scientific question.
- ▶ Use EDA to guide mean and variance structure.
- ▶ Choose:
 - ▶ Covariance pattern (direct correlation modeling), or
 - ▶ Linear mixed model (random effects).
- ▶ Check diagnostics at:
 - ▶ Mean level
 - ▶ Covariance level
 - ▶ Subject level
- ▶ Interpret in context and report sensitivity analyses.

Longitudinal modeling is iterative and model-based, not mechanical.